

Supplemental Material

Supplemental Material 1. SPIRIT 2025 Checklist of Items to Address in a Randomized Trial Protocol*

Section / Topic	No	SPIRIT 2025 checklist item description	Reported on page no.
Administrative information			
Title and structured summary	1a	Title stating the trial design, population, and interventions, with identification as a protocol	1
	1b	Structured summary of trial design and methods, including items from the World Health Organization Trial Registration Data Set	2 SM 2
Protocol version	2	Version date and identifier	6, 20
Roles and responsibilities	3a	Names, affiliations, and roles of protocol contributors	1, 23
	3b	Name and contact information for the trial sponsor	1
	3c	Role of trial sponsor and funders in design, conduct, analysis, and reporting of trial; including any authority over these activities	23
	3d	Composition, roles, and responsibilities of the coordinating site, steering committee, endpoint adjudication committee, data management team, and other individuals or groups overseeing the trial, if applicable	17
Open science			
Trial registration	4	Name of trial registry, identifying number (with URL), and date of registration. If not yet registered, name of intended registry	2, 6
Protocol and statistical analysis plan	5	Where the trial protocol and statistical analysis plan can be accessed	6
Data sharing	6	Where and how the individual de-identified participant data (including data dictionary), statistical code, and any other materials will be accessible	6, 7
Funding and conflicts of interest	7a	Sources of funding and other support (e.g., supply of drugs)	23
	7b	Financial and other conflicts of interest for principal investigators and steering committee members	23
Dissemination policy	8	Plans to communicate trial results to participants, healthcare professionals, the public, and other relevant groups (e.g., reporting in trial registry, plain language summary, publication)	11, 20
Introduction			

Background and rationale	9a	Scientific background and rationale, including summary of relevant studies (published and unpublished) examining benefits and harms for each intervention	4-6
	9b	Explanation for choice of comparator	4-6
Objectives	10	Specific objectives related to benefits and harms	6
Methods: Patient and public involvement, trial design			
Patient and public involvement	11	Details of, or plans for, patient or public involvement in the design, conduct, and reporting of the trial	20
Trial design	12	Description of trial design including type of trial (e.g., parallel group, crossover), allocation ratio, and framework (e.g., superiority, equivalence, non-inferiority, exploratory)	7, 11 Figure 2
Methods: Participants, interventions, and outcomes			
Trial setting	13	Settings (e.g., community, hospital) and locations (e.g., countries, sites) where the trial will be conducted	7-8
Eligibility criteria	14a	Eligibility criteria for participants	8-10 SM 3
	14b	If applicable, eligibility criteria for sites and for individuals who will deliver the interventions (e.g., surgeons, physiotherapists)	8
Intervention and comparator	15a	Intervention and comparator with sufficient details to allow replication including how, when, and by whom they will be administered. If relevant, where additional materials describing the intervention and comparator (e.g., intervention manual) can be accessed	11-13
	15b	Criteria for discontinuing or modifying allocated intervention/comparator for a trial participant (e.g., drug dose change in response to harms, participant request, or improving/worsening disease)	7, 13, 17 Figure 2
	15c	Strategies to improve adherence to intervention/comparator protocols, if applicable, and any procedures for monitoring adherence (e.g., drug tablet return, sessions attended)	16
	15d	Concomitant care that is permitted or prohibited during the trial	8-9 SM 3
Outcomes	16	Primary and secondary outcomes, including the specific measurement variable (e.g., systolic blood pressure), analysis metric (e.g., change from baseline, final value, time to event), method of aggregation (e.g., median, proportion), and time point for each outcome	13-15 SM 5 SM, Table 1
Harms	17	How harms are defined and will be assessed (e.g., systematically, non-systematically)	17

Participant timeline	18	Time schedule of enrollment, interventions (including any run-ins and washouts), assessments, and visits for participants. A schematic diagram is highly recommended (see Figure)	18-19 Table 1
Sample size	19	How sample size was determined, including all assumptions supporting the sample size calculation	19
Recruitment	20	Strategies for achieving adequate participant enrollment to reach target sample size	10-11
Methods: Assignment of interventions			
Randomization:			
Sequence generation	21a	Who will generate the random allocation sequence and the method used	11
	21b	Type of randomization (simple or restricted) and details of any factors for stratification. To reduce predictability of a random sequence, other details of any planned restriction (e.g., blocking) should be provided in a separate document that is unavailable to those who enroll participants or assign interventions	11
Allocation concealment mechanism	22	Mechanism used to implement the random allocation sequence (e.g., central computer/telephone; sequentially numbered, opaque, sealed containers), describing any steps to conceal the sequence until interventions are assigned	11
Implementation	23	Whether the personnel who will enroll and those who will assign participants to the interventions will have access to the random allocation sequence	11
Blinding	24a	Who will be blinded after assignment to interventions (e.g., participants, care providers, outcome assessors, data analysts)	11
	24b	If blinded, how blinding will be achieved and description of the similarity of interventions	11
	24c	If blinded, circumstances under which unblinding is permissible, and procedure for revealing a participant's allocated intervention during the trial	12
Methods: Data collection, management, and analysis			
Data collection methods	25a	Plans for assessment and collection of trial data, including any related processes to promote data quality (e.g., duplicate measurements, training of assessors) and a description of trial instruments (e.g., questionnaires, laboratory tests) along with their reliability and validity, if known. Reference to where data collection forms can be accessed, if not in the protocol	12-13, 15-16, 18 SM, Table 1 and 2
	25b	Plans to promote participant retention and complete follow-up, including list of any outcome data to be collected for participants who discontinue or deviate from intervention protocols	16
Data management	26	Plans for data entry, coding, security, and storage, including any related processes to promote data quality (e.g., double data entry; range checks for data values). Reference to	11

		where details of data management procedures can be accessed, if not in the protocol	
Statistical methods	27a	Statistical methods used to compare groups for primary and secondary outcomes, including harms	19-20
	27b	Definition of who will be included in each analysis (e.g., all randomized participants), and in which group	19-20
	27c	How missing data will be handled in the analysis	20
	27d	Methods for any additional analyses (e.g., subgroup and sensitivity analyses)	N/A
Methods: Monitoring			
Data monitoring committee	28a	Composition of data monitoring committee (DMC); summary of its role and reporting structure; statement of whether it is independent from the sponsor and funder; conflicts of interest and reference to where further details about its charter can be found, if not in the protocol. Alternatively, an explanation of why a DMC is not needed	N/A
	28b	Explanation of any interim analyses and stopping guidelines, including who will have access to these interim results and make the final decision to terminate the trial	17, 20
Trial monitoring	29	Frequency and procedures for monitoring trial conduct. If there is no monitoring, give explanation	18-19 Table 1
Ethics			
Research ethics approval	30	Plans for seeking research ethics committee/institutional review board approval	20
Protocol amendments	31	Plans for communicating important protocol modifications to relevant parties	20
Consent or assent	32a	Who will obtain informed consent or assent from potential trial participants or authorized proxies, and how	9, 18 SM4
	32b	Additional consent provisions for collection and use of participant data and biological specimens in ancillary studies, if applicable	N/A
Confidentiality	33	How personal information about potential and enrolled participants will be collected, shared, and maintained in order to protect confidentiality before, during, and after the trial	9, 11
Ancillary and post-trial care	34	Provisions, if any, for ancillary and post-trial care, and for compensation to those who suffer harm from trial participation	N/A

SM, Supplemental Material

*It is strongly recommend reading this checklist in conjunction with the SPIRIT 2025 Explanation and Elaboration and the SPIRIT 2025 Expanded Checklist for important clarifications on all the items. We also recommend reading relevant SPIRIT extensions. See www.consort-spirit.org

Supplemental Material 2. World Health Organization Trial Registration Data Set

Items		Information registered
1	Primary Registry and Trial Identifying Number	ClinicalTrials.gov, NCT06778122
2	Date of Registration in Primary Registry	January 10, 2025
3	Secondary Identifying Numbers	UCL - Project ID: 22105/001
4	Source(s) of Monetary or Material Support	IQAir AG (Steinach, Switzerland), PremaLabs Diagnostics, Ltd. (London, UK) & Lansion Biotechnology, Co. Ltd. (Nanjing, China) and A&D Instruments, Ltd. (Abingdon, UK)
5	Primary Sponsor	UCL
6	Secondary Sponsor(s)	N/A
7	Contact for Public Queries	Name: Eva Hernandez-Garcia; phone number: +44 (0)20 3108 3220; email: eva.hernandez.20@ucl.ac.uk
8	Contact for Scientific Queries	Name: Eva Hernandez-Garcia, Andrew Edkins; phone number: +44 (0)20 3108 3220; email: eva.hernandez.20@ucl.ac.uk, andrew.edkins@ucl.ac.uk
9	Public Title	In-Bedroom Renewed Air as Anti-inflammatory Adjuvant Therapy in Cancer Survivors (BREATHS)
10	Scientific Title	A series of N-of-1 randomised, adaptive, blinded, placebo-controlled trials of overnight in-bedroom air filtration as adjuvant treatment in reducing inflammatory biomarkers among survivors of adult-onset cancer with elevated CRP
11	Countries of Recruitment	Spain
12	Health Condition(s) or Problem(s) Studied	Cancer survivors, cardiotoxicity
13	Intervention(s)	Device: High efficiency particulate air filtration system Device: Air filtration system without filters (sham)
14	Key Inclusion and Exclusion Criteria	Inclusion: adult cancer survivors (breast, colorectal, rectum, prostate, non-small cell lung, Hodgkin/non-Hodgkin lymphoma, chronic lymphoid or myeloid leukaemia); age ≥ 18 years at diagnosis; ≤ 75 years at study entry; both sexes, all racial/ethnic backgrounds; non-smokers (≥ 12 months) and living in smoke-free households; completed definitive cancer treatment ≥ 12 months prior; on stable maintenance therapy ≥ 3 months if applicable; in complete remission; no metastatic disease; CRP ≥ 3 to <10 mg/L, or ≥ 10 mg/L if acute

		inflammation excluded; stable hypertension permitted (no med changes in prior 30 days; systolic <160 mmHg); other controlled chronic conditions permitted; permanent residence ≥3 months in high-IIPP (>125) Valencia neighbourhoods (Russafa, Malilla, Arrancapins); no plans to move; internet (Wi-Fi) access at home; signed informed consent. Additional: high or very high cardiovascular toxicity risk (per ESC Cardio-Oncology guidelines); prior cardiovascular disease, multiple risk factors, cardiotoxic cancer therapies (e.g., anthracyclines, chest radiotherapy, tyrosine kinase inhibitors, immune checkpoint inhibitors), or treatment-related cardiovascular complications. Exclusion: recent cancer diagnosis or planned cancer therapy/surgery during trial; metastatic disease; planned new prescription medications during trial; uncontrolled hypertension (systolic >160 mmHg over 10 days); unstable chronic disease within 6 months (e.g., uncontrolled diabetes); psychiatric illness impairing consent or compliance; chronic/recurrent infections or immunocompromised state (e.g., HIV); alcohol or substance misuse within past 12 months (including >14 drinks/week); shift work outside 07:00–22:00 hours; plans to relocate during trial or unwilling to spend ≥7 h overnight in bedroom; existing home air filtration system (unless disconnected during trial); refusal of informed consent.
15	Study Type	Interventional model: crossover assignment. Description: N-of-1 trial design (multiple crossover trial design for single participants). N-of-1 series (aggregated N-of-1 trial). Masking: triple (ParticipantCare, Provider, Investigator). Description: the sequence allocation will be masked for the participants, general practitioners, primary care staff and research team members, except for the field researcher to dispense the correct treatment.
16	Date of First Enrollment	Jan 23, 2026
17	Sample Size	10
18	Recruitment Status	Recruiting
19	Primary Outcome(s)	Change in blood CRP, measured weekly using portable point-of-care capillary microsampling (5 µL per participant; range 0.48–200 mg/L) at baseline and on days 7 and 14 of each period (between 07:00 and 10:00 am.)
20	Key Secondary Outcomes	Weekly point-of-care capillary measurements of Serum Amyloid A (5 µL; 2–300 µg/mL), and HbA1c (5 µL; 3–14%), as well as self-measured systolic and diastolic

		blood pressure (mean of three upper-arm oscillometric readings at 60-sec intervals).
21	Ethics Review	Approved. Research Ethics Committees: UCL (code: 22105/001), V.2.2 Amended Protocol Version; Doctor Peset University Hospital, Valencia-Spain (CEIm, code: 044/25); General University Hospital of Valencia, Valencia-Spain (CEIm, code: 1/2026 EO-PS).
22	Completion Date	Study Completion (Estimated): Jul 31, 2026
23	Summary Results	Results will be posted once available
24	IPD sharing statement	Yes: Identifiable participant information will not be shared. Pseudonymised data, with unique random ID codes and identifiers stored separately, will be accessible to study collaborators in the UK and outside the European Economic Area for research and manuscript preparation. Data will be stored and shared securely via the UCL REDCap project repository and Send-It application over encrypted connections.

Supplemental Material 3. Full Eligibility Criteria

Inclusion criteria:

1. Adult-onset cancer survivors with a confirmed histopathological diagnosis of cancer at any anatomical site (International Classification of Diseases, 10th Revision [ICD-10] codes C00-C97): breast (C50), colon (C18), colorectum (C19), rectum (C20), prostate (C61), NSCL (C34), Hodgkin lymphoma (C81), non-Hodgkin lymphoma (C82-C85), chronic lymphoid leukaemia (C91.1) and chronic myeloid leukaemia (C92.1) cancer.
2. Aged ≥ 18 years at cancer diagnosis—upper age limit to study entry = 75 years.
3. People of both genders, all racial and ethnic backgrounds.
4. Non-smokers (i.e., not smoking during previous 12 months).
5. A history of local, regional or distant recurrence, primary second cancers or multiple primary tumors will be not reasons to exclude.
6. No previous evidence for metastatic disease.
7. Evidence of complete remission defined as a decrease in or disappearance of signs and symptoms of cancer after treatment.
8. Having completed definitive treatment (surgery, neoadjuvant/adjuvant cytotoxic chemotherapy, radiotherapy, immunotherapy) at least 12 months to avoid acute inflammatory effects of therapy.^{15,16,99,100,17,92–98} Based on treatment-related immunosuppression—no upper limit on time elapsed from cancer(s) diagnosis.
9. Continuous, maintenance therapy is considered if initiated ≥ 3 months prior to study treatment and prescribed for long-term, chronic use without interruption during the trial, or completed for at least ≥ 3 months before starting the trial.
10. Having a persistently high inflammatory cardiac risk profile at the initial screening defined as a C-reactive protein (CRP) measurement ≥ 3 to <10 mg/L,⁶⁶ or ≥ 10 mg/L if acute inflammation is ruled out.¹⁰¹
11. Having stable hypertension (if applicable): no medication changes in prior 30 days, systolic blood pressure (BP) <160 mm Hg and either <130 mm Hg without antihypertensive medications, or diagnosis of hypertension in medical records.

12. Clinical evidence of cardiovascular, respiratory and/or musculoskeletal disease, and/or other controlled and stable chronic medical conditions will not be reason of exclusion.
13. Any medical prescription(s) if routinely administered and continued through the trial period.
14. Living at one of the urban areas of the city of Valencia with the highest index impact of pollutant on population values (IIPP > 125): Russafa, Malilla and Arrancapins neighbourhoods.¹⁰²
15. Permanent residence in their usual home at least 3 months prior to the first collected blood sample in the first cycle and without plans to move out toward another dwelling during the trial study.
16. Living in a nonsmoking household.
17. Internet connection (Wi-Fi) in the house.
18. Signed informed consent prior to commencement of specific protocol procedures.

Exclusion criteria:

1. Recently diagnosed with cancer or plans for additional cancer therapy and/or surgery (e.g., curative or reconstructive) during the trial period.
2. Metastatic disease.
3. Planned medication prescription to start during the trial.
4. Current chronic anti-inflammatory drug prescriptions or other chronic drugs (e.g., antihypertensive drugs) will be no reason for exclusion if its use is continued for the duration of the trial period.
5. Having uncontrolled hypertension: participants with average systolic BP >160 mmHg over any 10-day period prior and during the study.
6. History of uncontrolled, unstable chronic disease as defined by clinical practice guidelines/local medical practices and documented in medical records—within past 6 months to screening visit(s) and during the study—including, but not limited to:

Cardiac/vascular condition:

- Uncontrolled hypertension (investigator-defined).

- Unstable coronary artery disease (CAD) presenting with acute coronary syndromes—ST-elevation myocardial infarction, non-ST elevation myocardial infarction, unstable angina—characterized by new or worsening symptoms, electrocardiogram changes, or troponin elevation.¹ Stable CAD includes chronic coronary syndromes (obstructive atherosclerotic CAD or ischemia with non-obstructive coronary arteries) with symptoms controlled by medications or revascularization >6 months before screening.²
- Unstable peripheral arterial disease with recent revascularization (<6 months of the screening visit), revascularization for asymptomatic/mildly symptomatic disease or restenosis, or acute limb ischemia, or clinically significant tissue loss (severe ischemic ulcers or gangrene).³
- Unstable cerebrovascular disease, including ischemic or hemorrhagic stroke, transient ischemic attack, >1 symptomatic stroke, or severe neurological disability within 6 months,⁴ National Institutes of Health Stroke Scale score ≥ 14 or minimal change (± 1 point) on two assessments in patients with recent stroke, indicating poor functional outcomes and need for long-term care.^{5 6}

Metabolic condition:

- Uncontrolled diabetes mellitus (DM): HbA1c >7% (short-duration, no cardiovascular disease [CVD]), >8% (elderly, longstanding DM), or >9% (elderly, multiple comorbidities).⁷

Pulmonary disease:

- Unstable chronic obstructive pulmonary disease (COPD) with post-bronchodilator (PB) forced expiratory volume in one second (FEV₁)/forced vital capacity ratio <0.70, PB FEV₁% <80%, high symptom burden (COPD Assessment Test ≥ 10 or Modified Medical Research Council dyspnea score ≥ 2 , and ≥ 2 moderate or ≥ 1 severe exacerbations in the prior year).^{8 9}
- Uncontrolled asthma: ≥ 3 indicators of partially controlled asthma in any week, such as reduced FEV1 or peak expiratory flow (<80% predicted/best), frequent symptoms or rescue inhaler use (>2 days/month), nighttime awakenings, activity limitation, ≥ 1 systemic corticosteroid-treated exacerbation/year, or abnormal validated questionnaire scores.¹⁰

Musculoskeletal/rheumatologic disorder:

- Uncontrolled rheumatoid arthritis with moderate-to-severe disease activity (the most recent assessment at screening), defined by clinical disease activity index (CDAI) >10, disease activity score in 28 joint counts (CRP or Erythrocyte Sedimentation Rate) >3.2, or simplified disease activity index > 11. Patients in remission or with mild/low disease activity are considered stable.^{11 12}

Renal disease:

- Progressive chronic kidney disease defined on the basis of a declining estimated glomerular filtration rate (eGFR) and moderately increased proteinuria at screening: G5 (eGFR <15 mL/min/1.73 m²) or dialysis-dependent, G4 (eGFR <30 mL/min/1.73 m²) with progression/renal replacement need, or progressive G1–G3 (eGFR ≥30 mL/min/1.73 m²) with >5 mL/min/1.73 m² annual decline, >25% drop in 5 years, or proteinuria >1.0 g. Includes polycystic kidney disease (<50 y) and glomerular hematuria with proteinuria >0.5 g requiring potential immunosuppression.^{13 14}

Gastrointestinal disease:

- Active ulcerative colitis: Adapted Mayo score 5–9, with endoscopic subscore 2–3; includes stool frequency, rectal bleeding, and endoscopic mucosal appearance.
 - Active Crohn's disease: CDAI >220 within 6 months; endoscopic evidence of mucosal inflammation (≥6 for ileocolonic/colonic disease or ≥4 for isolated ileal disease); and either average daily stool frequency ≥4 or abdominal pain score ≥2.
7. Psychiatric illness or neurological disorders (e.g., Alzheimer's disease) that would limit compliance with trial requirements and/or prevent the patient from giving informed consent.
 8. History of ongoing, chronic or recurrent infectious disease, and having suspected or proven immunocompromised state (i.e., Human Immunodeficiency Virus infection).
 9. Self-report of alcohol or substance abuse within the past 12 months, including at-risk drinking (current consumption of more than 14 alcoholic drinks per week).
 10. Shift workers or subjects who have the work schedule falling outside the hours of 07:00 a.m. and 10:00 p.m.

11. Plans to move out of usual home during the trial period.
12. Unwilling to spend at least 7 hours/overnight in the bedroom.
13. Pre-existing an air filtration system that improves household air quality. If so, the device will be disconnected during all the trial cycles.
14. Unwilling to give consent.

Additional inclusion criteria linked to high or very high CV toxicity risk factors.

According to the European Society of Cardiology guidelines on cardio-oncology:¹⁵

1. High or very high baseline CV toxicity risk pre-treatment:
 - Prior history of CVD (e.g., heart failure [HF], venous thromboembolism [VTE], CAD).
 - Presence of multiple medical CV-related factors (e.g., DM, hypertension, dyslipidemia).
 - Current or concomitant cancer treatment (e.g., anthracycline before human epidermal growth factor receptor-type 2 [HER2]-targeted therapy; anthracycline and HER2-targeted therapy delivered concurrently).
 - Prior cardiotoxic cancer therapy (e.g., anthracycline, radiotherapy [RT] to left chest or mediastinum).
 - Lifestyle risk factors (history of smoking, body mass index [BMI] >30).
2. Specific cardiotoxic cancer therapies with a high risk of long-term CV complications:
 - Total lifetime cumulative dose anthracycline ≥ 250 mg/m² doxorubicin equivalent.
 - High-dose anthracycline ≥ 250 mg/m² doxorubicin equivalent.
 - High-dose RT > 15 Gray (Gy) Mean Heart Dose (MHD).
 - Anthracycline ≥ 100 mg/m² doxorubicin equivalent in combination with RT 5-15 Gy MHD.
 - Multitargeted tyrosine kinase inhibitors (TKI) targeting BCR-ABL and concomitant high-dose dexamethasone therapy >160 mg/month.
 - Combination of Rapidly Accelerated Fibrosarcoma and Mitogen-Activated Protein Kinase Kinase inhibition.
 - Single-agent immune checkpoints inhibitors (ICI) therapy (anti-PD1, e.g., nivolumab) followed by previous cardiotoxic cancer therapies: RT, anthracyclines, 5-fluorouracil, and TKI.

- Combination of two types of ICI (anti-PD1, e.g., nivolumab and anti-CTLA-4, e.g., ipilimumab).
3. Moderate or severe cancer therapy-related CV toxicity detected during cancer treatment or new symptomatic/asymptomatic abnormalities at the end of cancer treatment (3 or 12 months after therapy).

Supplemental Material 4. Informed Consent Form

Consent Form for Survivors of Adult-onset Cancer in Research Study

Please complete this form after you have read the Information Sheet and/or listened to an explanation about the research.

Trial Title: A series of n-of-1 randomised, adaptive, blinded, placebo-controlled trials of overnight in-bedroom air filtration as adjuvant treatment in reducing inflammatory biomarkers among survivors of adult-onset cancer with elevated CRP

Short Title: In-Bedroom Renewed Air as Anti-inflammatory Adjuvant Therapy in cancer Survivors (BREATHS) trial

UCL Research Ethics Committee Approval ID Number: 22105/001

Local Research Ethics Committee Approval ID Number: _____

Thank you for considering taking part in this research. The person organising the research must explain the project to you before you agree to take part. If you have any questions arising from the Information Sheet or explanation already given to you, please ask the researcher before you decide whether to join in. You will be given a copy of this Consent Form to keep and refer to at any time.

I confirm that I understand that by ticking each box below I am consenting to this element of the study. I understand that it will be assumed that unticked boxes means that I DO NOT consent to that part of the study. I understand that by not giving consent for any one element that I may be deemed ineligible for the study.

- | | Tick
Box |
|--|--------------------------|
| 1 I confirm that I have read and understood the Information Sheet for the above study. I have had an opportunity to consider the information and what will be expected of me. I have also had the opportunity to ask questions, which have been answered to my satisfaction. | ☐ |
| 2 I understand that my participation is voluntary, and I am free to withdraw at any time, without giving any reason, and without my medical care or legal rights being affected. | ☐ |
| 3 I understand that my personal information will be used for the purposes explained to me. I understand that according to data protection legislation, "public task" for personal data and "research purposes" for special category data will be the lawful basis for processing. Personal information collected in the study: <ul style="list-style-type: none">• Personal identifiers: name and surnames, date of birth, gender, full home addresses, telephone numbers, and email addresses.• Special category data: ethnicity and health data – medical history, treatments, examination data, test results, data from medical devices and health-related self-reported questionnaires. | ☐ |
| 4 I understand that if I choose to withdraw, data already collected will continue to be used for the purposes explained to me. | ☐ |

- 5 I understand the steps that have been taken to keep my personal details confidential, and that such information will be stored on a secure, passwords-protected platform. I understand that all efforts will be made to ensure I cannot be identified unless there are compelling and legitimate reasons for this to be breached. Research data will not be presented at any point with real names and will be pseudonymised by keeping the direct identifiers and the data separate, so that linkage to an identity is no longer possible without the additional information that is held separately. ☐
- 6 I understand that my information may be subject to review by responsible individuals from the University College London and/or regulatory authorities for monitoring and audit purposes. I permit these individuals access to my research records. ☐
- 7 I understand that the data will not be made available to any commercial organisations but is solely the responsibility of the researcher(s) undertaking this study. ☐
- 8 I give permission for the researcher to examine my medical records to get the information required for this study. ☐
- 9 I consent to my GP being informed of my participation within the study, and I understand that the researcher will contact my GP about my medical records to verify or complete full or part of medical data required in this study. ☐
- 10 I agree to provide 45 microlitres (µl) capillary blood sampling each week in the home visits. I understand that additional blood samples may be collected for repeated analysis in case of failure error in the medical device occurs. I understand that these microsamples will be collected following the procedures explained in the Information Sheet. I understand that the screening tests for CRP levels are being performed solely to determine eligibility for the study. ☐
- 11 I hereby confirm that I understand the inclusion and exclusion criteria as detailed in the Information Sheet and explained to me by the researcher, and that I do not fall under the exclusion criteria. ☐
- 12 I understand the possible disadvantages and risks of participating and the support that will be available to me should I become distressed during the research. ☐
- 13 I understand that there is no guarantee of benefits of participating. No promise of benefit has been made to encourage you to participate. ☐
- 14 I understand that I will not be monetarily compensated for the portion of time spent in the study and I will not benefit financially from any possible outcome it may result in the future. ☐
- 15 I understand that my participation will entail additional costs in the electrical bill due to the portable air filtration unit and air quality monitors in operation. I agree to take charge of these expenses during my participation in the study as detailed in the Information Sheet. ☐
- 16 I understand that the information collected will be published in scientific journals and presented at conferences. It will not be possible to identify me in any publications and presentation. ☐

- 17 I consent to being contacted by the researcher for the purposes of trial and I understand that this will require me to provide my contact details to the researcher. ☐
- 18 I agree that my GP may be contacted if any unexpected results are found in relation to my health. ☐
- 19 I wish to be contacted by e-mail or posted with a numerical and graphical report of my anonymised individual data, in relation to the results of the research, once the study is complete and discuss my results with the researcher. Your e-mail and delivery address will not be used for any other purpose. ☐
- 20 I wish to be contacted by e-mail with a copy of the publications of the research findings. Your e-mail address will not be used for any other purpose. ☐
- 21 I know how to contact the research team if I need to, and I am aware of who I should contact if I wish to lodge a complaint. ☐
- 22 I understand that the personal information will be securely stored during the trial and for 15 years after the trial completion. I understand that the personal data will not be used in future related projects. ☐
- 23 I understand that my anonymised research data may be shared to other study collaborators within the United Kingdom, and outside the European Economic Area for the research purposes. No one will be able to identify you when this data is shared. ☐
- 24 I agree to take part in the above named study. ☐

Name of participant

__ / __ / ____
Date

Signature

Researcher

__ / __ / ____
Date

Signature

Supplemental Material 5. Rationale for Selection of Secondary Biomarkers

1. **Serum amyloid A (SAA).** Circulating SAA has been shown to be a complementary predictor of CV risk in cancer survivors.¹⁶ Systemic inflammation, as measured by CRP and SAA at a later time point (after treatment), is a long-term prognostic factor regardless of age, tumor stage, BMI, and prior CV events.^{17 18} Evidence indicates that SAA can be additionally tested for detecting slight variations in systemic chronic inflammation, even when patients have a range of CRP values <3 mg/L,¹⁹ or when CRP variations are not detected.²⁰ Given its pro-inflammatory and pro-atherogenic properties, SAA test may be an early indicator of response to inflammation-lowering treatment linked to CV events.²¹
2. **Hemoglobin A1c (HbA1c).** A joint effect of mildly elevated HbA1c (6.0% to <6.5%) and CRP (> 3 mg/L) has been associated with a 2.5-fold increased risk of coronary heart disease among nondiabetic individuals compared to those with lower thresholds of both markers.²² Even modest elevations of HbA1c (>5.5%) and CRP (>1.8 mg/L) are linked to advanced subclinical carotid atherosclerosis progression and nearly a 3-fold increase in vascular events.²³
3. **Systolic and diastolic BP.** Individuals with elevated baseline BP, even in the pre-hypertensive range, are at increased risk for developing HF when treated with anthracycline.²⁴ Short-term BP variability, particularly in systolic BP, has been identified as an independent predictor of cancer therapy-related CV toxicity²⁵ and is associated with CV events including stroke and coronary events, even among individuals with normotensive or controlled BP levels.²⁶

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Table 1. Parameters of In-Vitro Rapid Quantitative Point-of-Care Testing Assays.

Test	Sample Volume	Reaction Time	Measuring Range
CRP	5µl	3 min	0.48-200 mg/L
SAA	5µl	3 min	2-300 µg/mL
HbA1c	5µl	5 min	3%-14%

Abbreviations: CRP, C-reactive protein; SAA, serum amyloid A; HbA1c, hemoglobin A1c; µL, microliters; µg/mL, micrograms per milliliter; min, minutes.

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Table 2. Participant Acceptability Questionnaire

Treatment with air filtration	
1	What were the barriers and enabling factors to completion of the intervention with air filtration devices?
2	Do you feel that you benefited from participating in the clinical trial?
3	Is there anything that you would add to or change about the intervention?
Time window for intervention	
4	Did the time window of the intervention (from 10 p.m. to 10 a.m.) meet your usual sleep patterns?
5	The intervention involved spending at least 7 consecutive hours in the bedroom each night. If your typical sleep duration was less than 7 hours, did you find it difficult to stay in the bedroom for the full 7 hours?
Data collection and methods	
Point-of-care blood testing at home & self-measured blood pressure monitoring	
6	What are your general thoughts on the method used for collecting blood microsamples at home? The intervention involved a finger-prick test using 45 microliters of capillary blood collected weekly with a point-of-care testing device.
7	What are your general thoughts about the method for blood pressure monitoring?
8	What were the advantages and disadvantages (e.g., barriers to usage, practical challenges) for you of having the test conducted at home?
9	Do you feel that having frequent and immediate access to test results could improve the quality of your care?
10	Now that you have experienced the blood collection process via rapid tests and point-of-care testing device, do you believe you can manage the process correctly and successfully take self-measurements of point-of-care blood tests at home (out-of-hours primary care)?
Self-reported questionnaires	
11	Do you think the questionnaires to be completed were manageable throughout the trial?
12	Were any of the questionnaires burdensome for you to complete weekly?

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